

Module 2: Attack Types and Threat Actors

Attack Types and CISSP Domains

Understanding various attack methods and how they map to the CISSP security domains is crucial for fortifying an organization against data breaches [cite: image_3b0b9b.png].

- **Password Attacks:** An attempt to access password-secured devices, systems, networks, or data [cite: image_3b0b9b.png]. Common forms include brute force and rainbow table attacks [cite: image_3b0b9b.png]. These fall under the Communication and Network Security domain [cite: image_3b0b9b.png].
- **Social Engineering Attacks:** A manipulation technique that exploits human error to gain private information, access, or valuables [cite: image_3b0b9b.png]. Forms include phishing, smishing, vishing, spear phishing, whaling, social media phishing, Business Email Compromise (BEC), watering hole attacks, USB baiting, and physical social engineering [cite: image_3b0b9b.png]. These are related to the Security and Risk Management domain [cite: image_3b0b9b.png].
- **Physical Attacks:** Security incidents that affect both digital and physical environments [cite: image_3b0b9b.png]. Examples include malicious USB cables, malicious flash drives, and card cloning/skimming [cite: image_3b0b9b.png]. These fall under the Asset Security domain [cite: image_3b0b9b.png].
- **Adversarial Artificial Intelligence:** A technique manipulating AI and machine learning technology to conduct attacks more efficiently [cite: image_3b0b9b.png]. This falls under both the Communication and Network Security and the Identity and Access Management domains [cite: image_3b0b9b.png].
- **Supply-Chain Attacks:** Targets systems, applications, hardware, and/or software to locate vulnerabilities [cite: image_3b0b9b.png]. Because every sold item involves third parties, breaches can happen at any point in the supply chain, affecting multiple organizations [cite: image_3b0b9b.png]. These span several domains, including Security and Risk Management, Security Architecture and Engineering, and Security Operations [cite: image_3b0b9b.png].
- **Cryptographic Attacks:** Affects secure forms of communication between a sender and intended recipient [cite: image_3b0b9b.png]. Forms include birthday, collision, and downgrade attacks [cite: image_3b0b9b.png]. These fall under the Communication and Network Security domain [cite: image_3b0b9b.png].

Threat Actor Types

Threat actors are defined by their malicious intent, and understanding their motivations helps organizations better prepare defenses [cite: image_465e22.png].

- **Advanced Persistent Threats (APTs):** These actors have significant expertise in accessing a network without authorization [cite: image_465e22.png]. APTs tend to research their targets (e.g., large corporations or government entities) in advance and remain undetected for extended periods [cite: image_465e22.png]. Their motivations include damaging critical infrastructure (like the power grid) or gaining access to intellectual property (such as trade secrets) [cite: image_465e22.png].
- **Insider Threats:** These individuals abuse their authorized access to obtain data that may harm an organization [cite: image_465e22.png]. Intentions often include sabotage, corruption, espionage, or unauthorized data access and leaks [cite: image_465e22.png].
- **Hacktivists:** Threat actors driven by a political agenda [cite: image_465e22.png]. They abuse digital technology to accomplish goals that may include demonstrations, propaganda, social change campaigns, or fame [cite: image_465e22.png].

Hacker Categories and Motivations

A hacker is any person who uses computers to gain access to computer systems, networks, or data [cite: image_465e22.png]. They are defined by their technical skills and motivations, falling into three main categories [cite: image_465e22.png]:

- **Authorized Hackers:** Also called ethical hackers, they follow a code of ethics and adhere to the law to conduct organizational risk evaluations [cite: image_465e22.png]. They are motivated to safeguard people and organizations from malicious actors [cite: image_465e22.png].
- **Semi-Authorized Hackers:** Considered researchers, they search for vulnerabilities but do not take advantage of them [cite: image_465e22.png].
- **Unauthorized Hackers:** Also known as unethical hackers, these are malicious threat actors who do not follow or respect the law [cite: image_465e22.png]. Their primary goal is to collect and sell confidential data for financial gain [cite: image_465e22.png].

Other Hacker Profiles

- **New and Unskilled Threat Actors:** Their goals include learning and enhancing skills, seeking revenge, or exploiting weaknesses using existing malware and programming scripts [cite: image_465e22.png].

- **Contracted Hackers:** Motivated solely by completing the job they were contracted to do, working on both illegal and legal tasks for pay [cite: image_465e22.png].
- **Vigilantes:** Hackers whose main goal is to protect the world from unethical hackers [cite: image_465e22.png].